

## VERHNEE PENZINO, Russian Federation

WMO: 255380

Lat: 64.217N

Lon: 164.233E

Elev: 1068

StdP: 14.14

Time Zone: 12.00 (E12)

Period: 86-16

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB   |              | Humidification DP/MCDB and HR |            |              |              |            |              | Coldest Month WS/MCDB |              |             |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|--------------|--------------|-------------------------------|------------|--------------|--------------|------------|--------------|-----------------------|--------------|-------------|-------------|-----------------------|------------|--------------|
|               |              |              | 99.6%                         |            |              | 99%          |            |              | 0.4%                  |              | 1%          |             |                       |            |              |
|               | 99.6%        | 99%          | DP                            | HR         | MCDB         | DP           | HR         | MCDB         | WS                    | MCDB         | WS          | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>-56.3 | (c)<br>-53.0 | (d)<br>-59.7                  | (e)<br>0.2 | (f)<br>-55.6 | (g)<br>-56.4 | (h)<br>0.2 | (i)<br>-53.0 | (j)<br>17.7           | (k)<br>-13.6 | (l)<br>15.2 | (m)<br>-9.6 | (n)<br>2.1            | (o)<br>350 | (p)<br>0.811 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 19.2                   | 76.1            | 59.4 | 71.9 | 57.4 | 68.1 | 55.7 | 61.1                | 72.6 | 59.0 | 69.6 | 56.7 | 66.3 | 4.0                  | 130  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 55.7                            | 68.7 | 65.9 | 53.7 | 63.9 | 63.8 | 51.8 | 59.4 | 60.9 | 27.7          | 72.4 | 26.2 | 70.0 | 24.7 | 66.3 |                   |
|                                 |      |      |      |      |      |      |      |      |               |      |      |      |      |      |                   |

## Extreme Annual Design Conditions

| Extreme Annual WS |          |          | Extreme Annual Temperature |          |                    |          | n-Year Return Period Values of Extreme Temperature |           |            |           |            |           |            |           |    |    |
|-------------------|----------|----------|----------------------------|----------|--------------------|----------|----------------------------------------------------|-----------|------------|-----------|------------|-----------|------------|-----------|----|----|
|                   |          |          | Mean                       |          | Standard Deviation |          | n=5 years                                          |           | n=10 years |           | n=20 years |           | n=50 years |           |    |    |
| 1%                | 2.5%     | 5%       | Min                        | Max      | Min                | Max      | Min                                                | Max       | Min        | Max       | Min        | Max       | Min        | Max       | DB | WB |
| (a) 16.9          | (b) 14.2 | (c) 12.2 | (e) -60.5                  | (f) 81.9 | (g) 5.1            | (h) 2.6  | (i) -64.2                                          | (j) 83.8  | (k) -67.2  | (l) 85.3  | (m) -70.1  | (n) 86.8  | (o) -73.8  | (p) 88.7  |    |    |
| (4) 16.9          | (5) 14.2 | (6) 12.2 | (7) -60.5                  | (8) 81.9 | (9) 5.1            | (10) 2.6 | (11) -64.2                                         | (12) 83.8 | (13) -67.2 | (14) 85.3 | (15) -70.1 | (16) 86.8 | (17) -73.8 | (18) 88.7 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 14.4   | -24.5 | -15.9 | -7.1  | 8.5   | 32.6  | 50.5  | 55.6  | 48.9  | 36.5  | 14.5  | -7.1  | -21.3 |
|                                                                                  | DBStd       | 31.39  | 17.03 | 19.64 | 15.56 | 12.02 | 7.55  | 6.86  | 5.85  | 6.32  | 7.44  | 14.21 | 16.96 | 18.96 |
|                                                                                  | HDD50       | 13325  | 2308  | 1845  | 1771  | 1244  | 540   | 76    | 17    | 94    | 405   | 1099  | 1714  | 2212  |
|                                                                                  | HDD65       | 18464  | 2773  | 2265  | 2236  | 1694  | 1005  | 437   | 297   | 498   | 854   | 1564  | 2164  | 2677  |
|                                                                                  | CDD50       | 342    | 0     | 0     | 0     | 0     | 0     | 90    | 190   | 61    | 1     | 0     | 0     | 0     |
|                                                                                  | CDD65       | 5      | 0     | 0     | 0     | 0     | 0     | 1     | 4     | 0     | 0     | 0     | 0     | 0     |
|                                                                                  | CDH74       | 154    | 0     | 0     | 0     | 0     | 0     | 42    | 95    | 17    | 0     | 0     | 0     | 0     |
|                                                                                  | CDH80       | 9      | 0     | 0     | 0     | 0     | 0     | 3     | 5     | 1     | 0     | 0     | 0     | 0     |
| Wind                                                                             | WSAvg       | 3.6    | 3.6   | 3.7   | 3.8   | 4.1   | 3.6   | 3.8   | 3.7   | 3.4   | 3.5   | 3.2   | 3.7   | 3.7   |
| Precipitation                                                                    | PrecAvg     | 12.7   | 0.9   | 0.6   | 0.6   | 0.5   | 0.5   | 1.2   | 1.7   | 1.7   | 1.3   | 1.0   | 1.3   | 0.9   |
|                                                                                  | PrecMax     | 21.0   | 2.8   | 1.7   | 1.7   | 1.8   | 1.5   | 3.3   | 4.6   | 4.0   | 4.6   | 2.3   | 6.6   | 2.2   |
|                                                                                  | PrecMin     | 7.8    | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.2   | 0.4   | 0.4   | 0.0   | 0.2   | 0.2   | 0.1   |
|                                                                                  | PrecStd     | 3.3    | 0.7   | 0.4   | 0.5   | 0.4   | 0.4   | 0.9   | 0.9   | 0.7   | 1.1   | 0.5   | 1.1   | 0.6   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 26.1  | 28.3  | 29.1  | 35.8  | 57.3  | 79.0  | 82.0  | 76.5  | 60.0  | 43.4  | 32.0  | 28.2  |
|                                                                                  |             | MCWB   | 24.3  | 25.5  | 27.6  | 31.5  | 45.2  | 58.5  | 62.3  | 61.2  | 48.5  | 39.0  | 30.7  | 26.3  |
|                                                                                  | 2%          | DB     | 16.2  | 25.1  | 25.6  | 32.6  | 51.7  | 74.5  | 77.7  | 70.0  | 54.8  | 39.2  | 28.4  | 21.1  |
|                                                                                  |             | MCWB   | 14.5  | 23.5  | 23.3  | 29.5  | 41.8  | 56.8  | 60.7  | 58.2  | 46.3  | 35.9  | 26.4  | 19.7  |
|                                                                                  | 5%          | DB     | 6.9   | 20.8  | 21.0  | 30.2  | 47.4  | 70.4  | 73.5  | 65.3  | 51.3  | 36.2  | 23.6  | 13.6  |
|                                                                                  |             | MCWB   | 5.6   | 19.5  | 19.1  | 27.5  | 39.0  | 55.0  | 58.8  | 55.5  | 44.9  | 33.7  | 22.4  | 12.8  |
|                                                                                  | 10%         | DB     | -0.4  | 14.7  | 16.3  | 26.9  | 43.3  | 66.3  | 69.8  | 61.3  | 48.5  | 32.8  | 18.1  | 6.8   |
|                                                                                  |             | MCWB   | -1.4  | 13.6  | 14.7  | 24.2  | 36.7  | 52.7  | 57.0  | 53.2  | 43.8  | 30.5  | 16.9  | 5.7   |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 24.1  | 26.1  | 27.9  | 32.1  | 45.5  | 61.3  | 64.8  | 62.4  | 49.7  | 39.6  | 30.9  | 26.1  |
|                                                                                  |             | MCDB   | 26.1  | 27.5  | 29.0  | 34.6  | 55.9  | 72.7  | 76.0  | 74.3  | 55.7  | 43.0  | 31.9  | 27.9  |
|                                                                                  | 2%          | WB     | 14.6  | 23.7  | 23.5  | 30.2  | 42.0  | 58.5  | 62.2  | 59.3  | 47.9  | 36.4  | 26.5  | 19.9  |
|                                                                                  |             | MCDB   | 16.1  | 25.3  | 25.4  | 32.7  | 51.0  | 71.4  | 74.6  | 68.6  | 52.7  | 38.5  | 28.1  | 21.2  |
|                                                                                  | 5%          | WB     | 5.9   | 19.4  | 19.3  | 27.3  | 39.3  | 55.9  | 60.2  | 56.5  | 46.2  | 33.8  | 22.4  | 13.0  |
|                                                                                  |             | MCDB   | 7.0   | 20.8  | 20.9  | 30.0  | 45.8  | 68.6  | 71.5  | 63.7  | 50.0  | 36.0  | 23.8  | 13.7  |
|                                                                                  | 10%         | WB     | -1.2  | 13.5  | 14.7  | 24.3  | 37.3  | 53.5  | 58.3  | 54.1  | 44.3  | 30.5  | 17.0  | 5.7   |
|                                                                                  |             | MCDB   | -0.3  | 14.7  | 16.3  | 26.8  | 42.6  | 65.9  | 68.4  | 60.3  | 48.0  | 32.6  | 18.1  | 6.6   |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 15.3  | 17.9  | 20.7  | 20.7  | 13.2  | 20.2  | 19.2  | 18.6  | 16.9  | 14.8  | 15.6  | 13.8  |
|                                                                                  |             | MCDBR  | 14.9  | 12.1  | 12.7  | 14.2  | 19.6  | 28.7  | 29.2  | 27.6  | 22.1  | 12.7  | 16.2  | 14.8  |
|                                                                                  |             | MCWBR  | 14.3  | 11.2  | 10.6  | 11.4  | 11.4  | 14.7  | 14.8  | 16.3  | 14.5  | 9.5   | 14.3  | 14.3  |
|                                                                                  | 5% WB       | MCDBR  | 15.0  | 11.8  | 11.0  | 13.3  | 17.4  | 26.0  | 25.5  | 23.6  | 17.5  | 11.3  | 15.5  | 15.2  |
|                                                                                  |             | MCWBR  | 14.3  | 11.1  | 9.9   | 10.7  | 10.6  | 14.3  | 14.4  | 14.4  | 12.6  | 8.5   | 14.3  | 14.8  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.166  | 0.236 | 0.296 | 0.355 | 0.401 | 0.375 | 0.447 | 0.426 | 0.311 | 0.262 | 0.183 | 0.165 |       |
|                                                                                  | taud        | 1.961  | 2.153 | 2.066 | 1.928 | 1.935 | 2.319 | 2.127 | 2.138 | 2.453 | 2.211 | 1.936 | 1.923 |       |
|                                                                                  | Ebn at Noon | 142    | 221   | 245   | 253   | 252   | 266   | 239   | 230   | 248   | 203   | 119   | 56    |       |
|                                                                                  | Edh at Noon | 8      | 18    | 31    | 46    | 50    | 36    | 42    | 38    | 22    | 17    | 8     | 3     |       |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 55     | 300   | 869   | 1590  | 1814  | 1618  | 1541  | 1142  | 689   | 327   | 90    | 20    |       |
|                                                                                  | RadStd      | 7      | 28    | 53    | 63    | 134   | 177   | 154   | 136   | 80    | 54    | 10    | 1     |       |

## Historical Trends

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                        |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (0 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page